

Bangladesh University of Professionals



Faculty of Science and Technology Department of Information & Communication Technology (ICT)

Course Title: Internship

Course Code: ICE-3211

Submitted by

Name: Mst. Labonno Rahman Oishy

ID: 2254901107

Walton ID: IA-540

Section: A

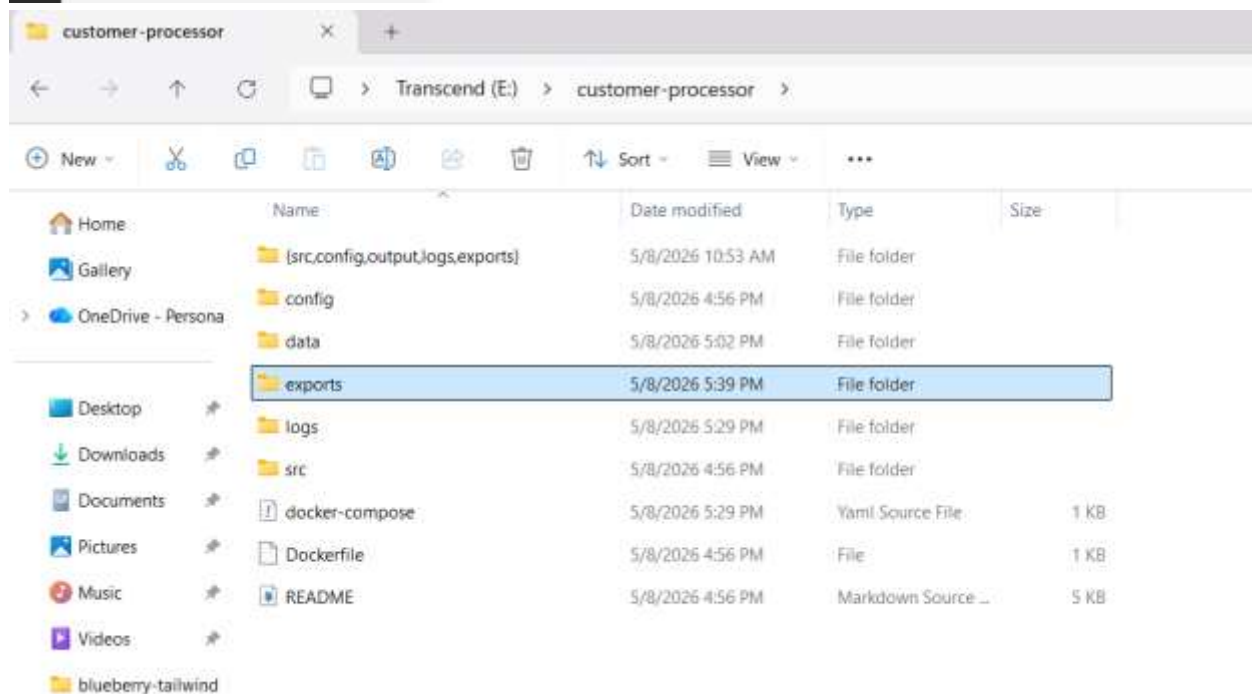
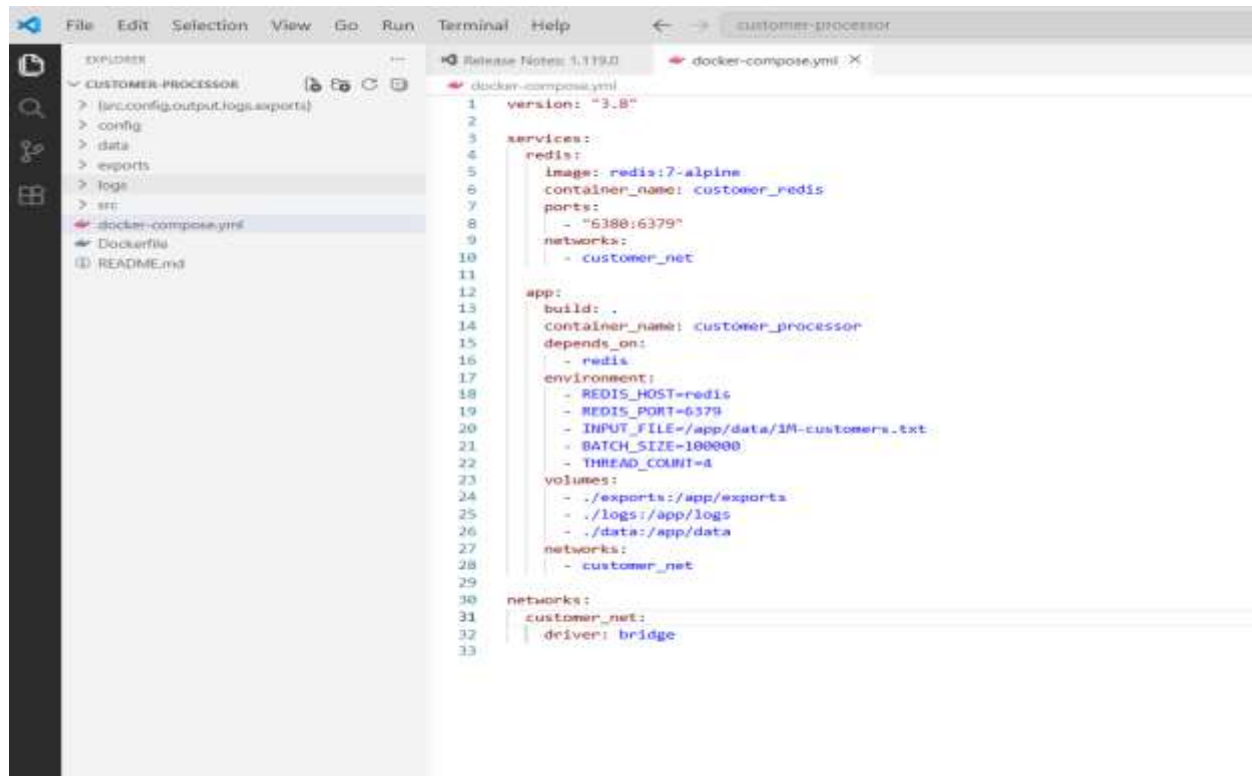
Batch: BICE – 2022

Session: 2021-22

Output:

[View in Docker Desktop](#) [View Config](#) [Enable Watch](#) [Getch...](#)





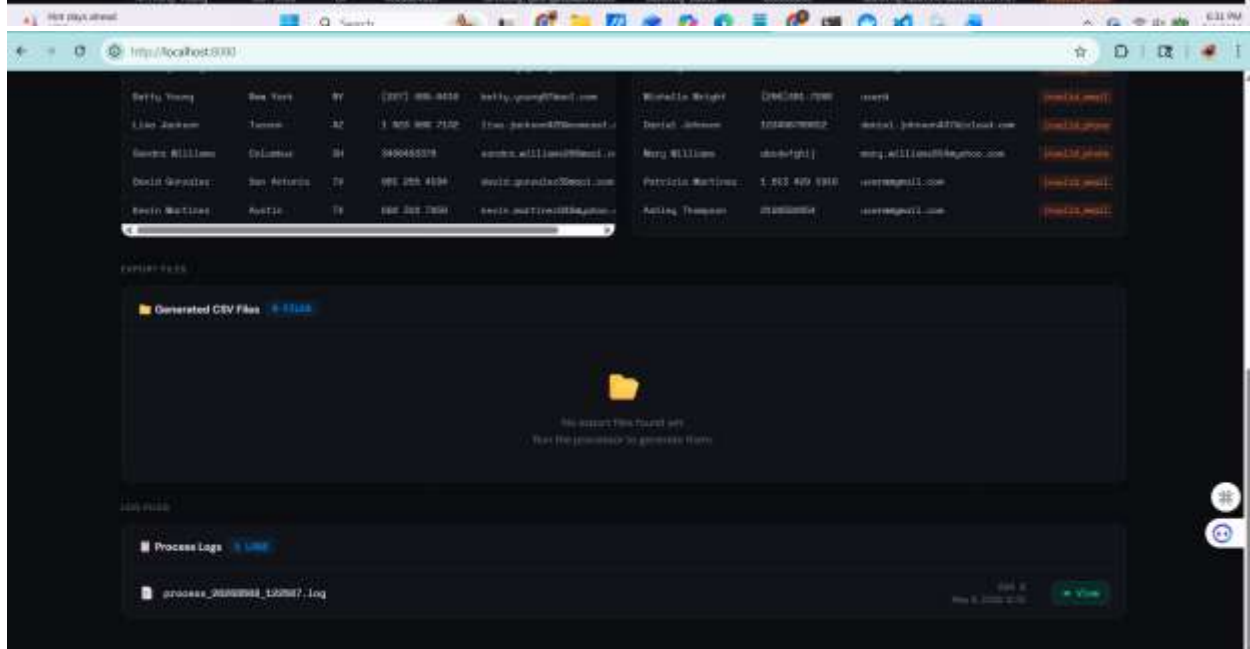
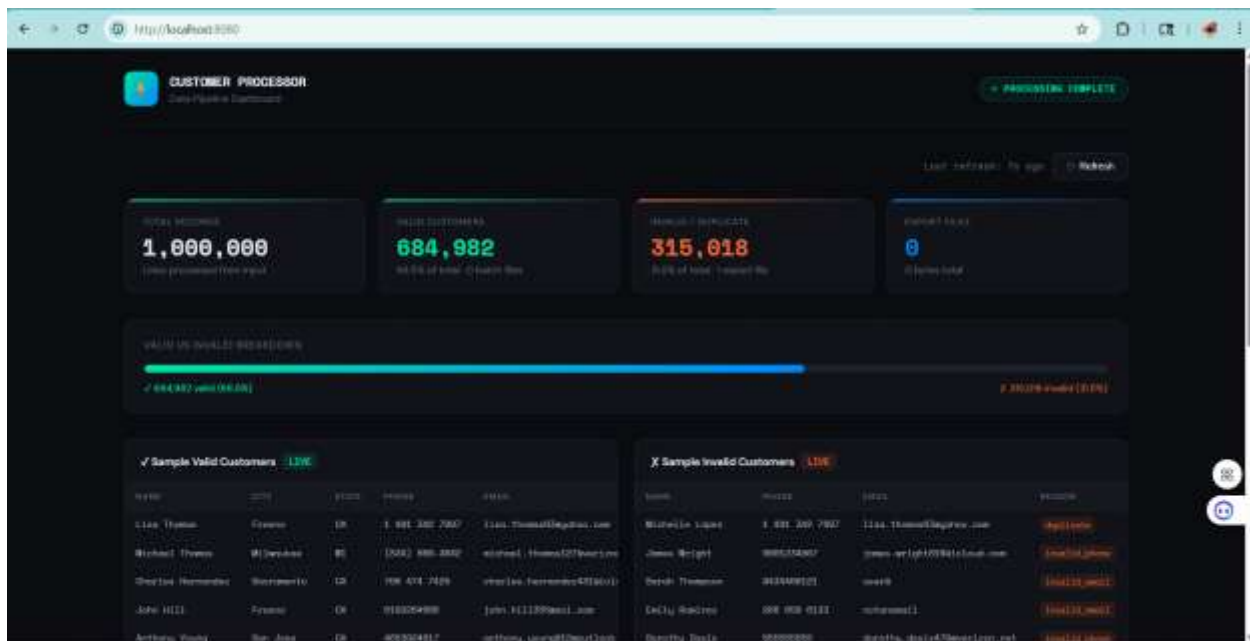
exports

Transcend (E:) > customer-processor > exports

New -

Sort - View -

	Name	Date modified	Type	Size
Home	invalid_customers	5/8/2026 5:40 PM	Microsoft Excel Co...	29,302 KB
Gallery	valid_customers_batch_1	5/8/2026 5:38 PM	Microsoft Excel Co...	8,593 KB
OneDrive - Persona	valid_customers_batch_2	5/8/2026 5:38 PM	Microsoft Excel Co...	8,595 KB
	valid_customers_batch_3	5/8/2026 5:39 PM	Microsoft Excel Co...	8,595 KB
Desktop	valid_customers_batch_4	5/8/2026 5:39 PM	Microsoft Excel Co...	8,595 KB
Downloads	valid_customers_batch_5	5/8/2026 5:39 PM	Microsoft Excel Co...	8,597 KB
Documents	valid_customers_batch_6	5/8/2026 5:39 PM	Microsoft Excel Co...	8,593 KB
Pictures	valid_customers_batch_7	5/8/2026 5:40 PM	Microsoft Excel Co...	7,303 KB
Music				
Videos				



Generated CSV Files 4 Files

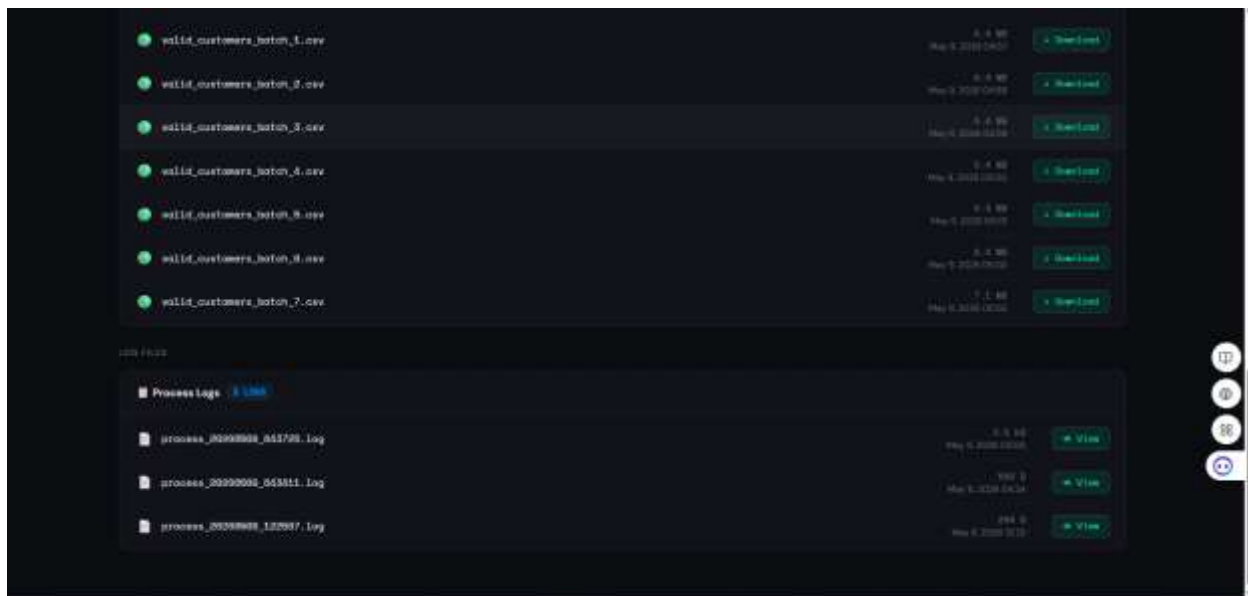
No export files found yet.
Run the processor to generate them.

Process Logs 1 Log

process_20250908_152827.log

Out: 8
In: 8, 2025-9-8

View



```
tch_7.csv
customer_processor | [2026-05-09 05:05:31] [INFO ] [PID:1] Exported 315018 invalid customers + /app/exports/invalid_c
customers.csv
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] =====
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] PROCESSING SUMMARY
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] =====
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] Total lines read : 1000000
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] Valid customers : 684982
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] Invalid customers : 315018
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] Malformed lines : 0
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] File processing time : 1180.34s
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] Export time : 506.4764s
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] Total execution time : 1686.9427s
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] =====
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] Export files:
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] [VALID] /app/exports/valid_customers_batch_1.csv
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] [VALID] /app/exports/valid_customers_batch_2.csv
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] [VALID] /app/exports/valid_customers_batch_3.csv
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] [VALID] /app/exports/valid_customers_batch_4.csv
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] [VALID] /app/exports/valid_customers_batch_5.csv
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] [VALID] /app/exports/valid_customers_batch_6.csv
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] [VALID] /app/exports/valid_customers_batch_7.csv
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] [INVALID] /app/exports/invalid_customers.csv
customer_processor | [2026-05-09 05:05:32] [SUCCESS] [PID:1] =====
customer_processor | [2026-05-09 05:05:32] [INFO ] [PID:1] Log file: /app/logs/process_20260509_043725.log
customer_processor | [2026-05-09 05:05:32] [INFO ] [PID:1] Done. Redis connection closed.
customer_processor exited with code 0
```

version: "3.8"

services:

redis:

image: redis:7-alpine

container_name: customer_redis

ports:

- "6380:6379"

networks:

- customer_net

app:

build: .

container_name: customer_processor

depends_on:

- redis

environment:

- REDIS_HOST=redis

- REDIS_PORT=6379

- INPUT_FILE=/app/data/1M-customers.txt

- BATCH_SIZE=100000

- THREAD_COUNT=4

volumes:

- ./exports:/app/exports

- ./logs:/app/logs

- ./data:/app/data

networks:

- customer_net

web:

build:

context: .

dockerfile: Dockerfile.web

container_name: customer_web

depends_on:

- redis

ports:

- "8080:80"

environment:

- REDIS_HOST=redis

- REDIS_PORT=6379

volumes:

- ./web:/var/www/html # Apache/Nginx

- ./exports:/app/exports

- ./logs:/app/logs

networks:

- customer_net

restart: unless-stopped

networks:

customer_net:

driver: bridge

<?php

/**

* CustomerValidator

*

- * Validates phone numbers and email addresses according to US formatting rules.

- * No third-party libraries are used — all logic is implemented with native PHP.

- *

- * Valid US phone formats (per spec):

- * 555-555-5555

- * (555)555-5555

- * (555) 555-5555

- * 555 555 5555

- * 5555555555

- * 1 555 555 5555

- */

```
class CustomerValidator
```

```
{
```

```
    /**
```

```
        * Validates a US phone number.
```

```
        *
```

```
        * Strategy:
```

```
        * 1. Strip all formatting characters (spaces, dashes, dots, parentheses).
```

```
        * 2. Strip a leading country code of "1" if the result is 11 digits.
```

```
        * 3. Ensure exactly 10 digits remain.
```

```
        * 4. Ensure the area code and exchange code do not start with 0 or 1
```

```
        * (NANP rules: area code NXX, exchange NXX where N = 2-9).
```

```
        *
```

```
        * @param string $phone Raw phone string from the input file.
```

```
        * @return bool
```

```
    */
```

```
public static function isValidPhone(string $phone): bool
{
    // Remove all non-digit characters
    $digits = preg_replace('/\D/', '', $phone);

    if ($digits === null) {
        return false;
    }

    $len = strlen($digits);

    // Accept 11-digit numbers only if they begin with the country code "1"
    if ($len === 11) {
        if ($digits[0] !== '1') {
            return false;
        }
        $digits = substr($digits, 1);
    }

    if (strlen($digits) !== 10) {
        return false;
    }

    // NANP: area code must not start with 0 or 1
    if ($digits[0] === '0' || $digits[0] === '1') {
        return false;
    }
}
```

```

    }

    // NANP: exchange code (positions 3-5) must not start with 0 or 1
    if ($digits[3] === '0' || $digits[3] === '1') {
        return false;
    }

    return true;
}

/**
 * Validates an email address.
 *
 * Uses PHP's built-in FILTER_VALIDATE_EMAIL which implements RFC 5321/5322
 * rules without any external library.
 *
 * @param string $email Raw email string from the input file.
 * @return bool
 */
public static function isValidEmail(string $email): bool
{
    if (empty(trim($email))) {
        return false;
    }

    return filter_var($email, FILTER_VALIDATE_EMAIL) !== false;
}

```

```
}

/**
 * Runs all validation rules on a Customer and populates its invalidReasons.
 *
 * @param Customer $customer Customer object to validate (mutated in place).
 */
public static function validate(Customer $customer): void
{
    if (!self::isValidPhone($customer->phone)) {
        $customer->invalidReasons[] = 'invalid_phone';
    }

    if (!self::isValidEmail($customer->email)) {
        $customer->invalidReasons[] = 'invalid_email';
    }
}
}
```